

Author Contribution Report

Distributed Node Health Monitoring Framework

June 25, 2026

1 Project Information

Paper Title: Distributed Node Health Monitoring Framework
Course: Distributed Systems
Supervisor: Dr. Tran Huu Tam
Submission Date: 25 Jun 2026

2 Authors

#	Name	Student ID
1	Nguyen Ha Khai	10423053
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4	Ninh Hoang Quan	10423097
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3 Contribution Matrix (CRediT)

Contribution Role	NHK	NDK	VMD	NHQ	THP
Conceptualization	P	S	S	P	S
Methodology	S	P	S	S	P
Software Development	P	P	S	P	S
Validation	S	S	–	S	P
Formal Analysis	P	P	P	–	–
Data Collection / Experiments	–	–	P	P	P
Visualization	S	S	P	S	P
Writing	P	P	P	P	P
Project Administration	P	–	–	P	S
Literature Review	–	P	P	–	–

Symbols are defined as follows:

- P : Primary Contributor
- S : Supporting Contributor
- – : No Contribution

4 Detailed Contributions

Author 1: Nguyen Ha Khai

Primary Roles

- Co-leader with Ninh Hoang Quan
- Algorithm analyst
- Manuscript Editor

Specific Contributions

- Implement FINDING NEMO & CHUALANH
- Write the FINDING NEMO component in implementation details and proposed approach section
- Write the system design part in related work section

Author 2: Nguyen Duy Khanh

Primary Roles

- Software engineer
- Algorithm designer

Specific Contributions

- Implement hybrid SWIMPHI
- Design CHUALANH algorithm
- Write the hybrid SWIMPHI component in implementation details and proposed approach section
- Write the failure detection part in related work section

Author 3: Vo Minh Duy

Primary Roles

- Frontend developer
- Literature reviewer

Specific Contributions

- Design the monitoring dashboard for data collection and system performance evaluation
- Review 90% citations in the research paper
- Research the architecture of the system
- Write the abstract, introduction, problem statement, and complete architectural diagram

Author 4: Ninh Hoang Quan

Primary Roles

- Backend developer
- Leader

Specific Contributions

- Implement the backbone or foundation of nodes in the system
- Implement the backend server monitoring application
- Contribute to the implementation of CHUALANH algorithm
- Write the CHUALANH algorithm in implementation details and proposed approach section
- Write Server Application part in the implementation details

Author 5: To Huynh Phuc

Primary Roles

- Manual tester
- Code reviewer
- Algorithm designer

Specific Contributions

- Design FINDING NEMO and hybrid SWIMPHI algorithm
- Write the evaluation; conclusion and future work sections
- Refactor the source code and ensure cohesive interactions between system components

5 Estimated Contribution Percentages

Author	Contribution (%)
Author 1	20
Author 2	20
Author 3	20
Author 4	20
Author 5	20
Total	100%

6 Declaration

We certify that the contribution information provided in this report accurately reflects each author's participation in the project and manuscript preparation.

Author Signature

Date
